

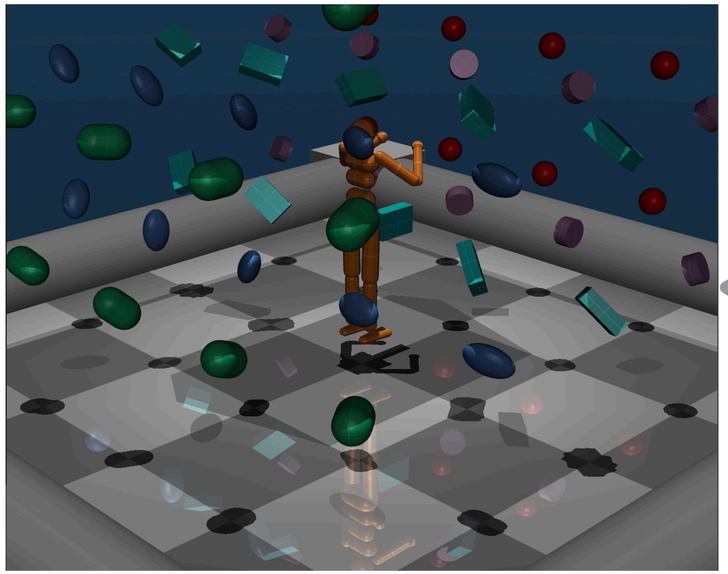
Point of Contact

Learning transferrable world models via contact supervision

Bilkit Githinji, Walter Talbott & Josh Susskind - September 29, 2022

Apple Confidential–Internal Use Only

Robot Learning & Data Scarcity



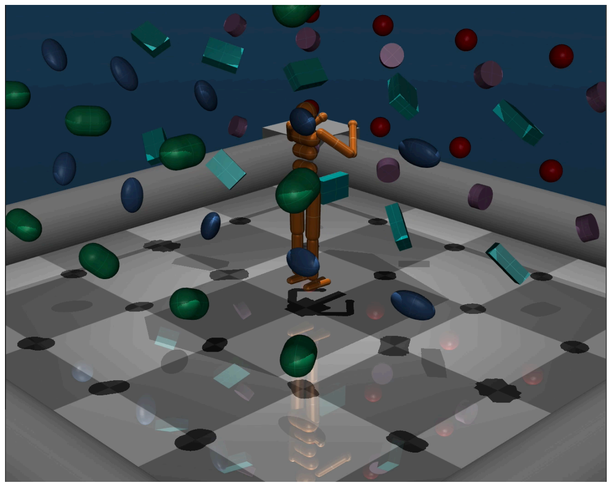
policy transfer?



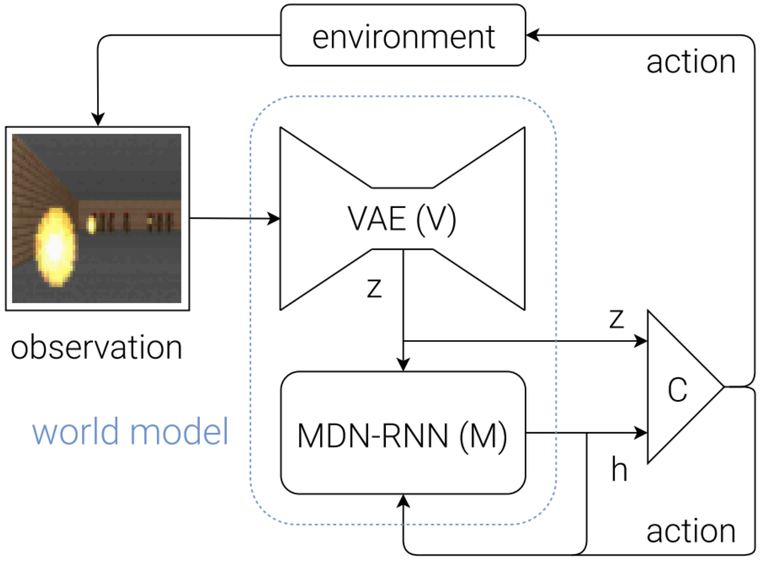
src: https://spectrum.ieee.org

Apple Confidential–Internal Use Only

Model-based Reinforcement Learning



Non-linear + stochastic dynamics, high-dimensional state, epistemic uncertainty



Ha & Schmidhuber (2018)

We consider a Partially Observable MDP

> image observations, unknown physical parameters

We prefer the model-based paradigm in our approach

> typically sample efficient, robust, transferrable, interpretable

Ha & Schmidhuber (2018); Vikash Kumar (2016); Todorov et al. (2012)

Apple Confidential–Internal Use Only

Contact-rich Continuous Control

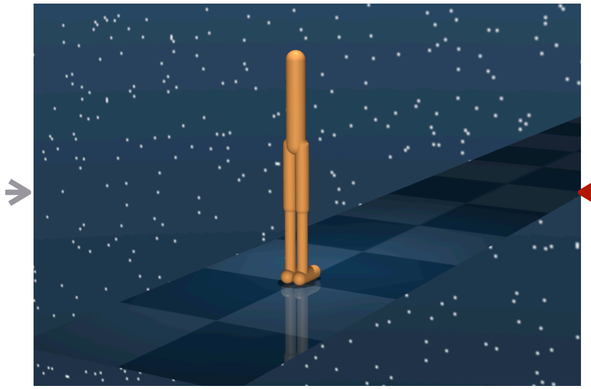
A key challenge for world models.

Vision Only

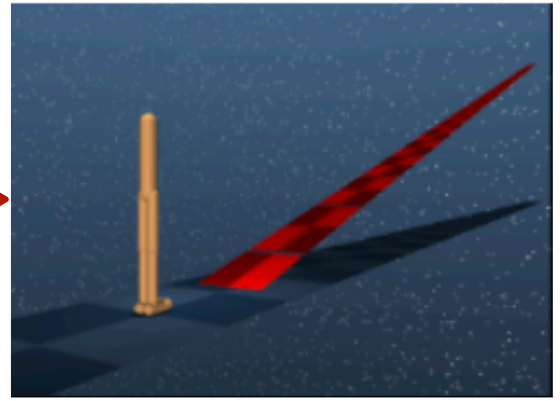


src: https://spectrum.ieee.org

Vision + Contacts



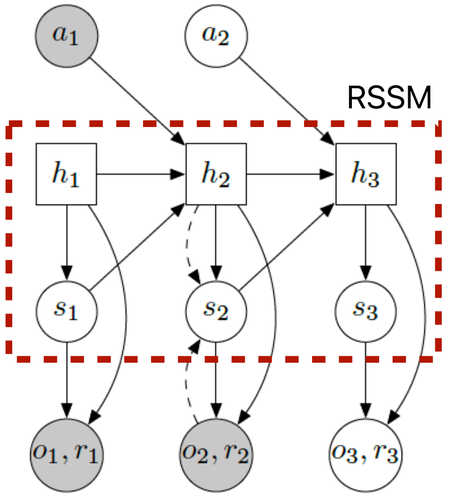
Vision Only



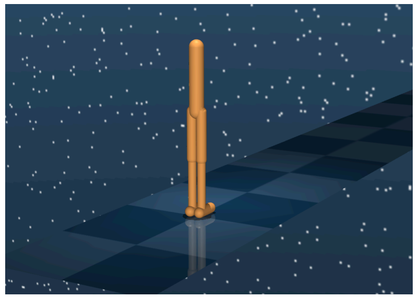
Underlying Question Can multi-modal supervision improve world model's test-time performance?

Apple Confidential–Internal Use Only

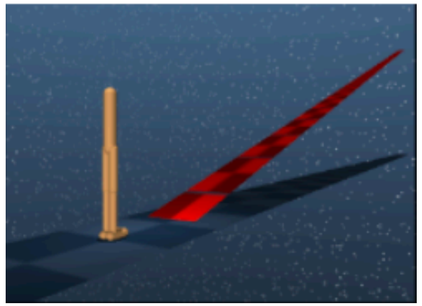
Baseline Test Performance



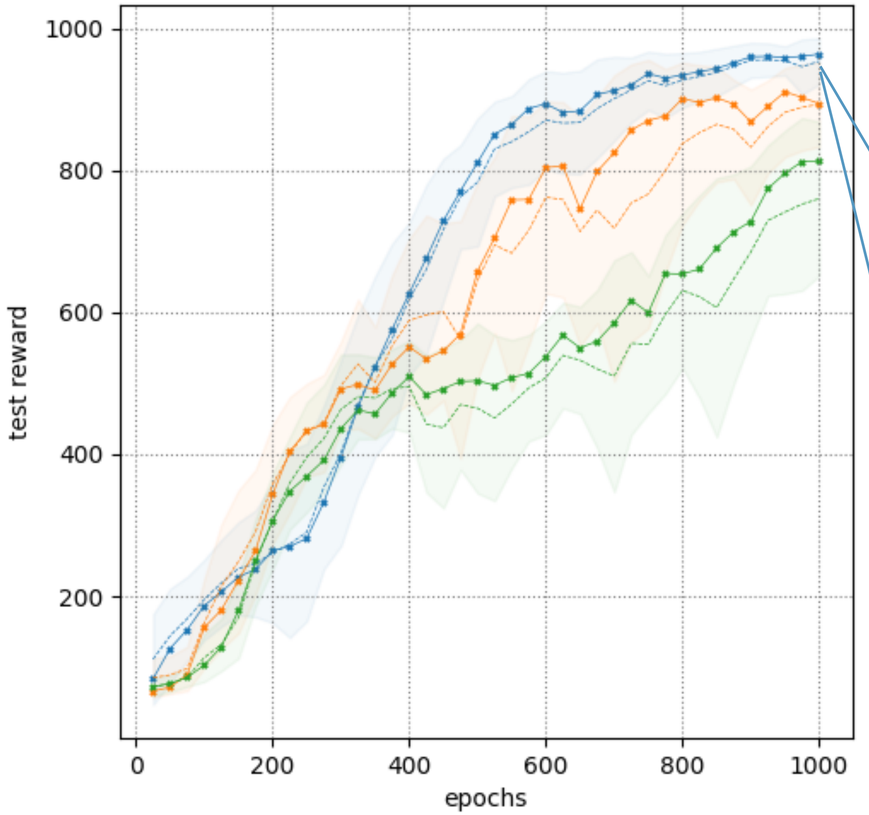
Hafner et al. (2018)



flat ground



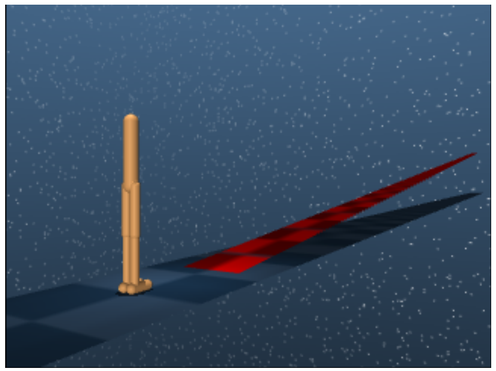
ramp



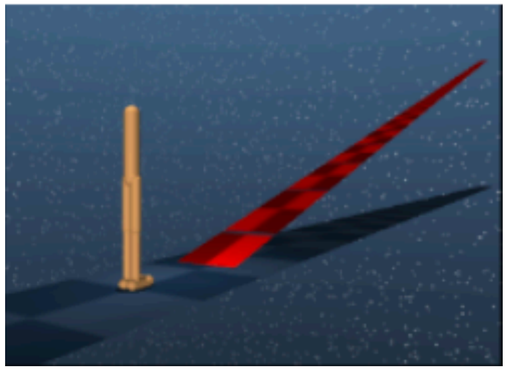


Apple Confidential–Internal Use Only

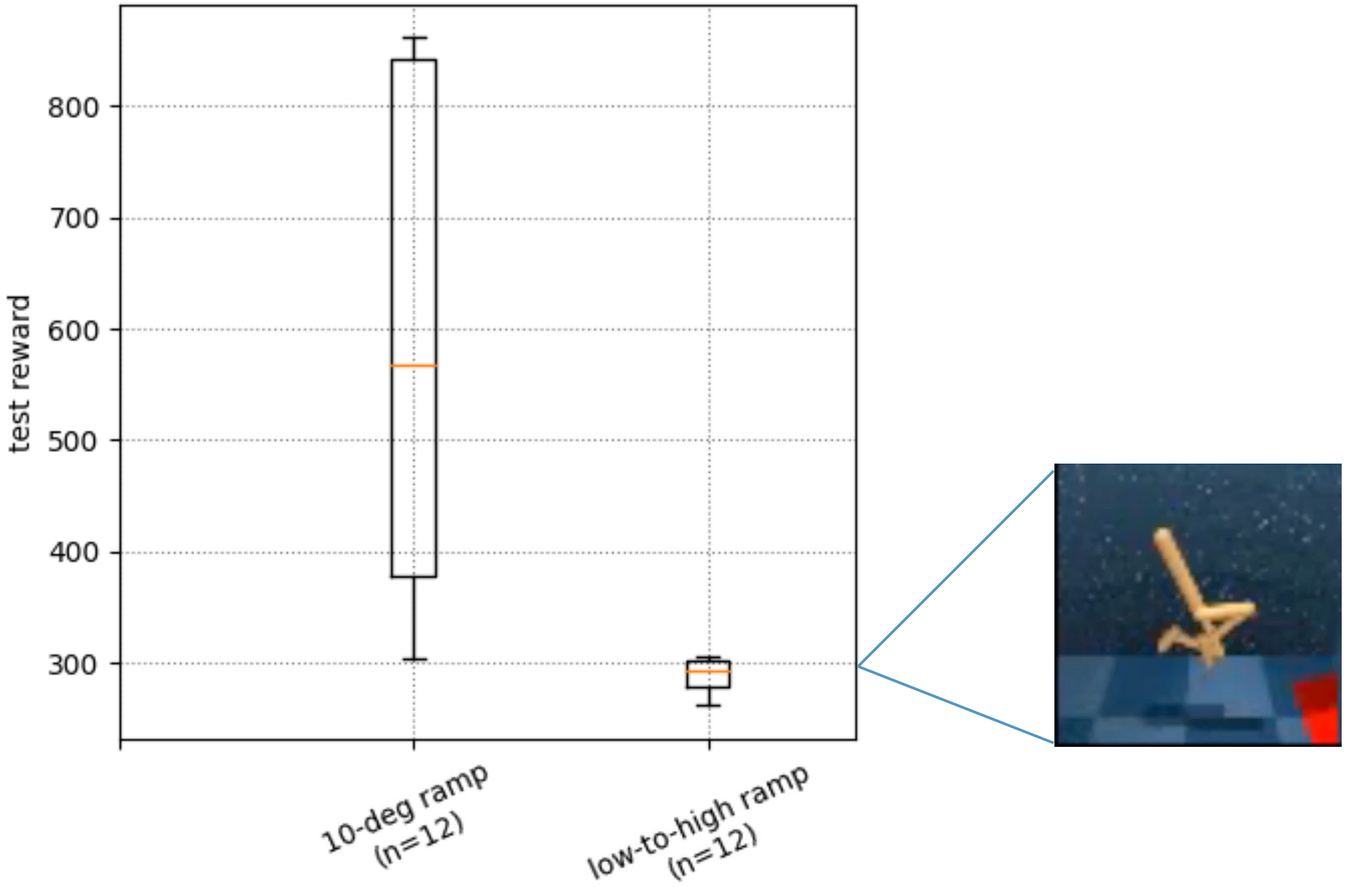
Baseline Generalisation Performance



low ramp

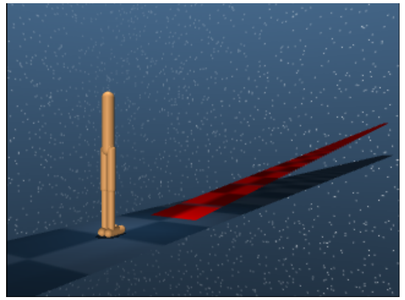


high ramp

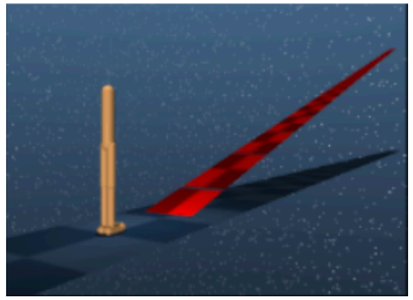


Apple Confidential–Internal Use Only

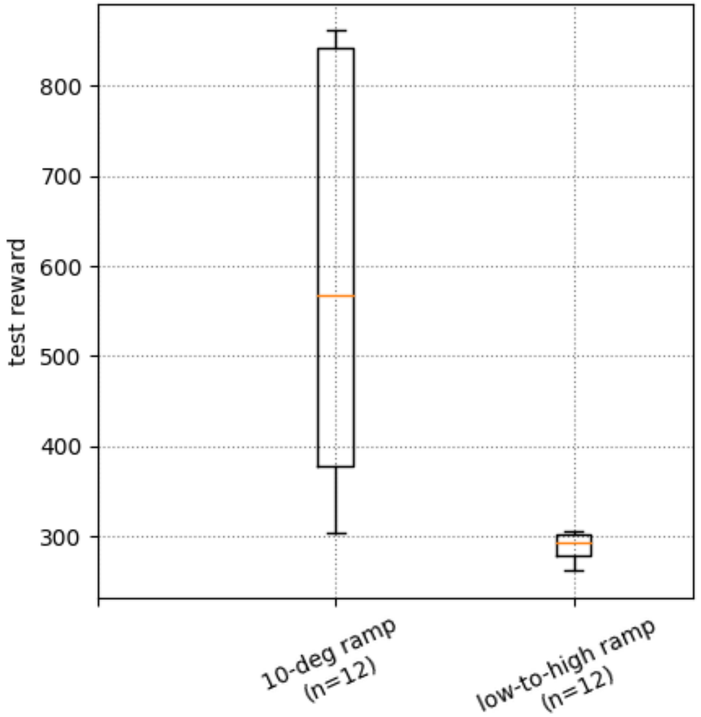
Baseline Generalisation Performance



low ramp



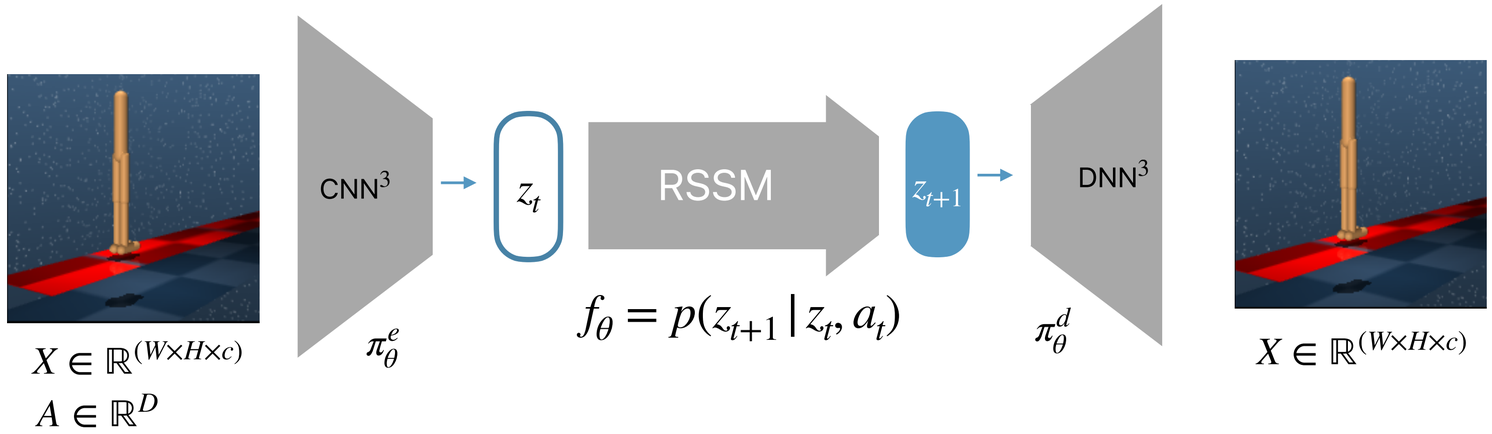
high ramp



Hypothesis Training world models with contact supervision will guide visual representations to encode more relevant info about physical state

Apple Confidential–Internal Use Only

Baseline World Model



Hafner et al. (2018)

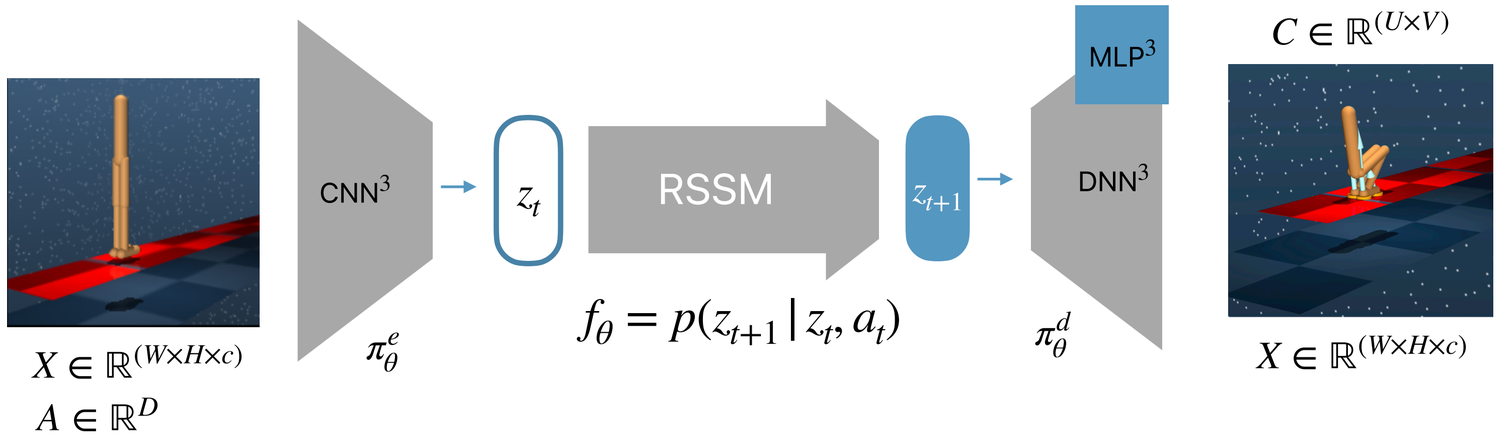
$R \in \mathbb{R}$ → MLP^3 $r_\theta = p(r_t \mid z_{t-1})$

Learn a world model $M_\theta = \{\pi_\theta^e, f_\theta, \pi_\theta^d, r_\theta\}$ that maximises the likelihood of observed trajectory $p(x \mid a)$. Then optimise a policy $\pi(x; M_\theta)$ to maximise expected return R.

Apple Confidential–Internal Use Only

Our Approach

A world model trained with contact supervision.

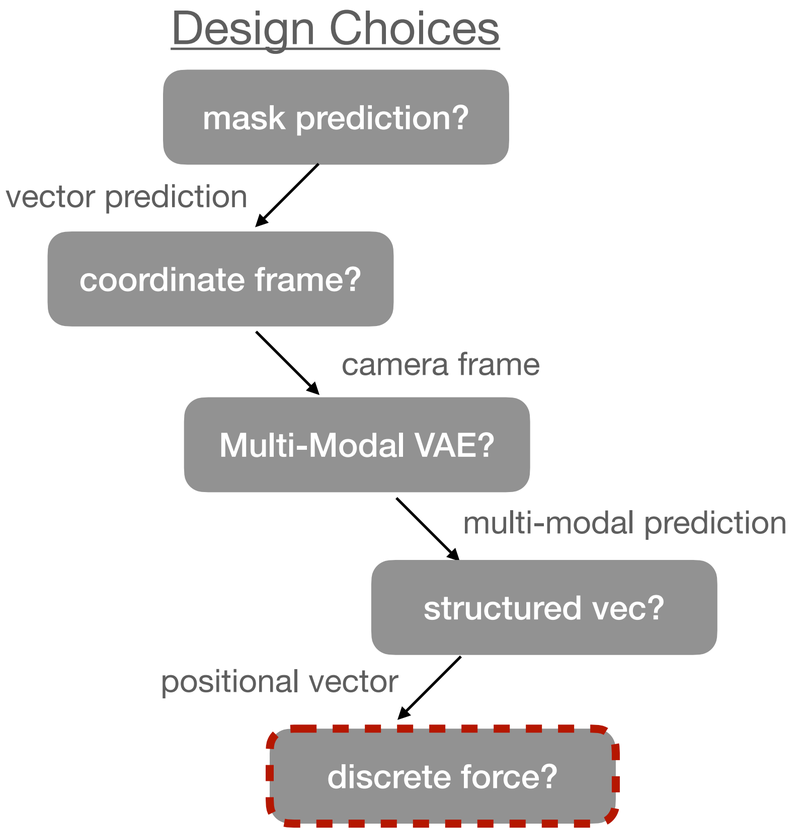


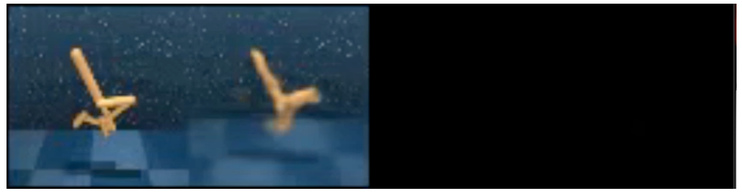
Learn a world model $M_\theta = \{\pi_\theta^e, f_\theta, \pi_\theta^d, r_\theta\}$ that maximises the likelihood of observed trajectory $p(x, c \mid a)$, assuming that $I(X; C) > \delta > 0$. Then optimise a policy $\pi(x; M_\theta)$ to maximise expected return R.

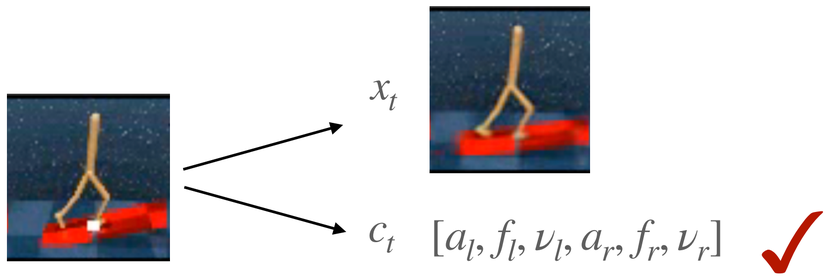
Apple Confidential–Internal Use Only

Representation of Contacts

What form of contact supervision yields higher performing policies?



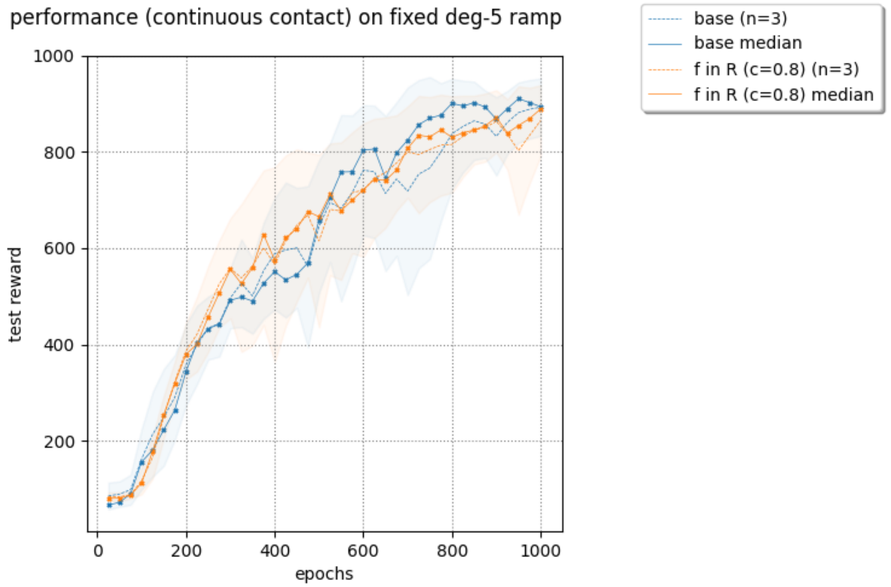




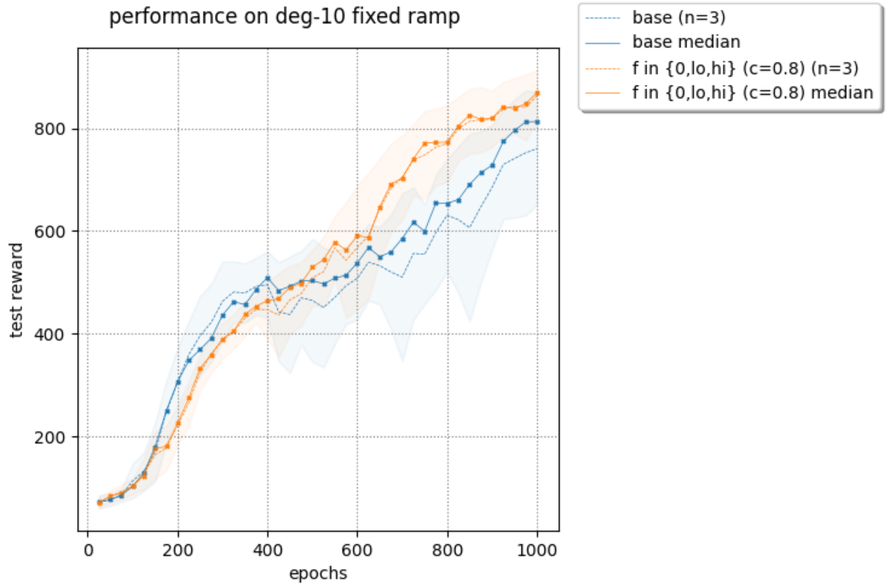
Apple Confidential–Internal Use Only

Structured Contact Vector

Supervision with a structured vector yields higher performing policies.



simplified contact
continuous contact force

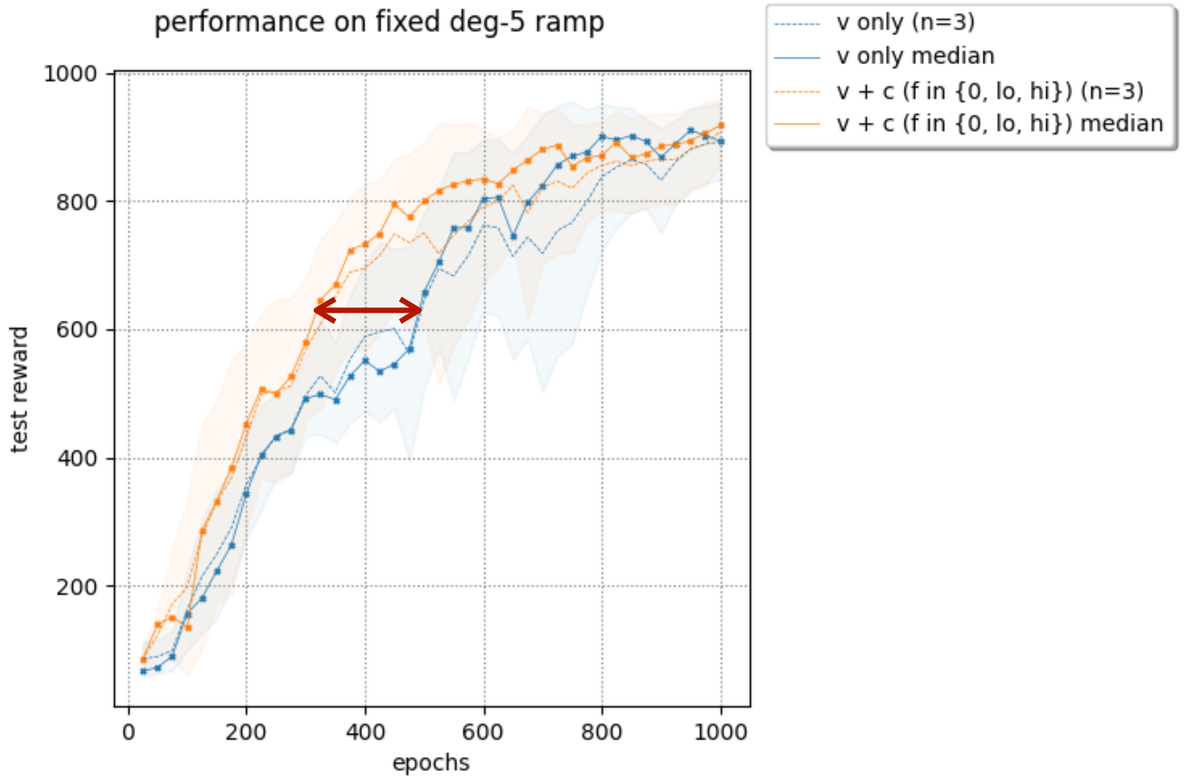


simplified contact
quantised contact force

Apple Confidential–Internal Use Only

Efficiency

Contact-aware world models yield policies that learn faster.

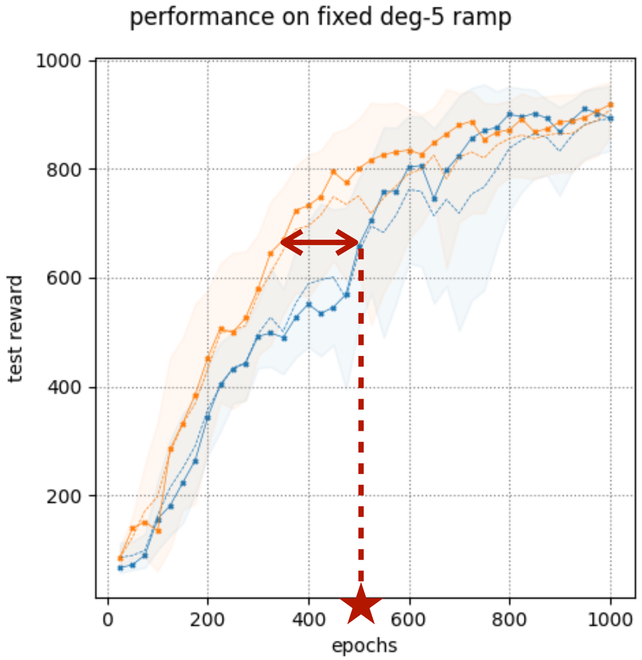


simplified contact
quantised contact force

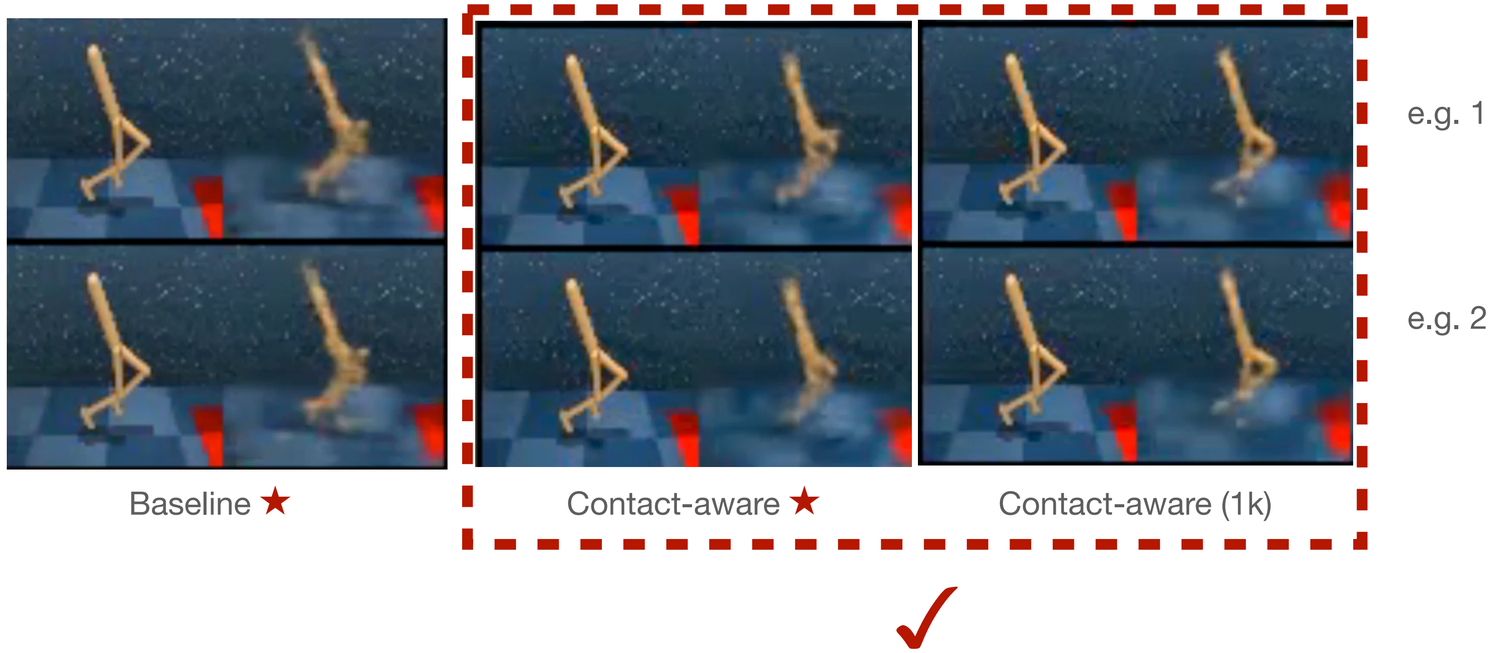
Apple Confidential–Internal Use Only

Efficiency

Contact-aware world models yield policies that learn faster.



simplified contact
quantised contact force



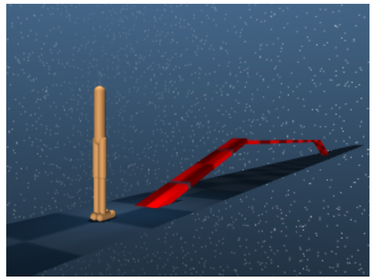
✓

Apple Confidential–Internal Use Only

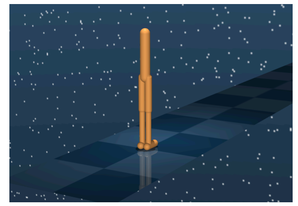
Generalisation

How robust are contact-aware world models to physical variations?

Case 1: generalisation across terrains

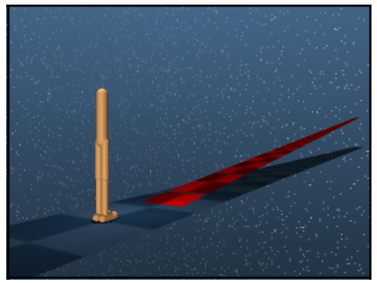


$\theta \sim \text{Unif}(10^\circ, 15^\circ)$

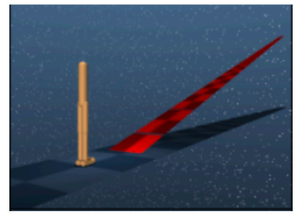


$\theta = 0^\circ$

Case 2: generalisation to unseen terrains



$\theta \sim \text{Unif}(0^\circ, 5^\circ)$

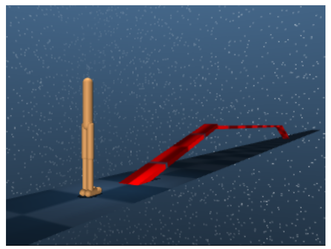


$\theta \sim \text{Unif}(5^\circ, 10^\circ)$

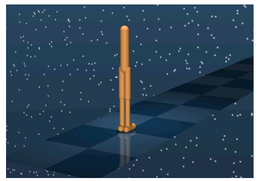
Apple Confidential–Internal Use Only

Key Finding

Contact-aware world models yield policies capable of *zero-shot transfer.*

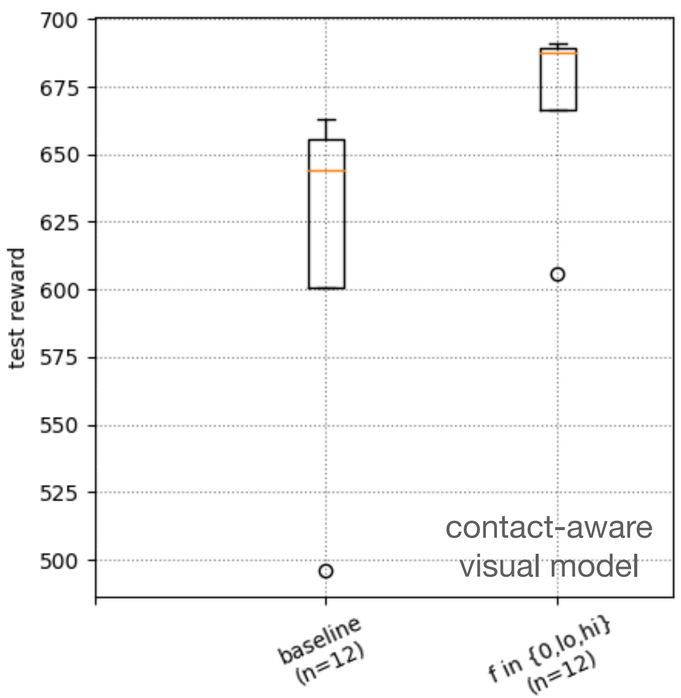


$\theta \sim \text{Unif}(10°,15°)$



$\theta = 0°$

Case 1: generalisation
across terrains



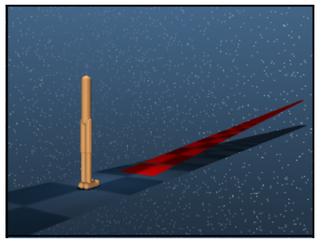


✓

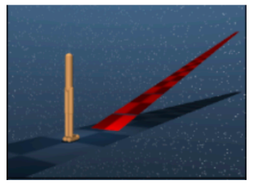
Apple Confidential–Internal Use Only

Key Finding

Contact-aware world models yield policies capable of *zero-shot transfer.*

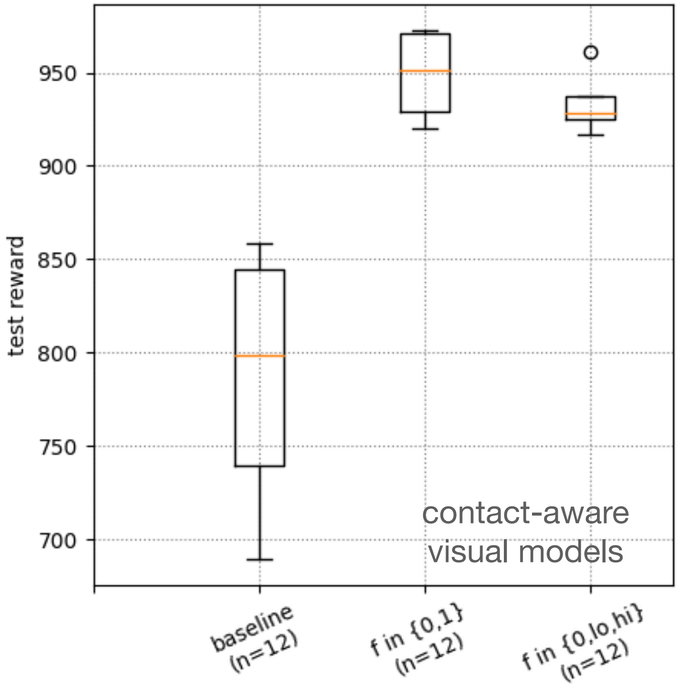


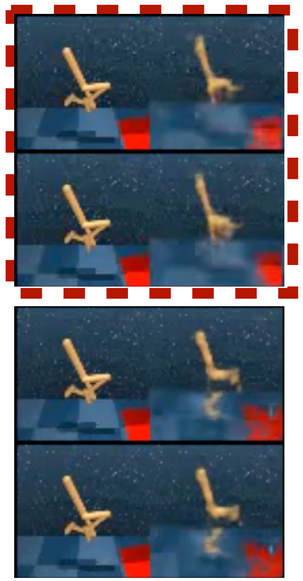
$\theta \sim \text{Unif}(0°,5°)$



$\theta \sim \text{Unif}(5°,10°)$

Case 2: generalisation to
unseen terrains





✓

Apple Confidential–Internal Use Only

Point of Contact

Continuing work

Adroit manipulation benchmark

- Tasks: door opening, in-hand manipulation, tool use, etc.
- SOTA uses demonstration guided model-free RL

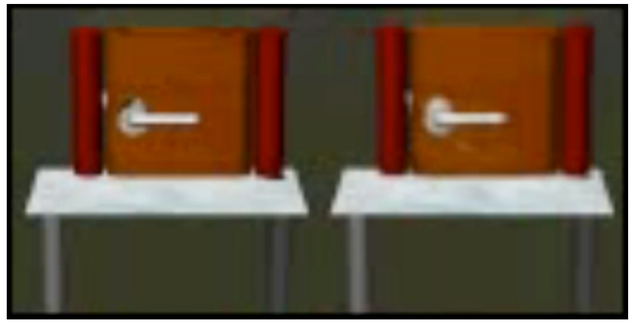
Modelling hybrid dynamics

- Apply more structure inspired by analytical models?
- Go beyond contact prediction task?

Adroit Door-V0 (**adapted**)



Rajeswaran et al. (2017)



Visual GT Prediction

Apple Confidential–Internal Use Only

Acknowledgements 🥹

Helpful Discussions Vimal Thilak, Devon Hjelm, Andrew Albright, Tudor Cristea-Platon, Shuangfei Zhai, Miguel Sarabia del Castillo, Rin Metcalf Susa, Nico Lingg

MLR Interns Dhruv Sreenivas, Enric Boix, Peiye Zhuang, Ayush Jain

Apple Interns Angela Liu

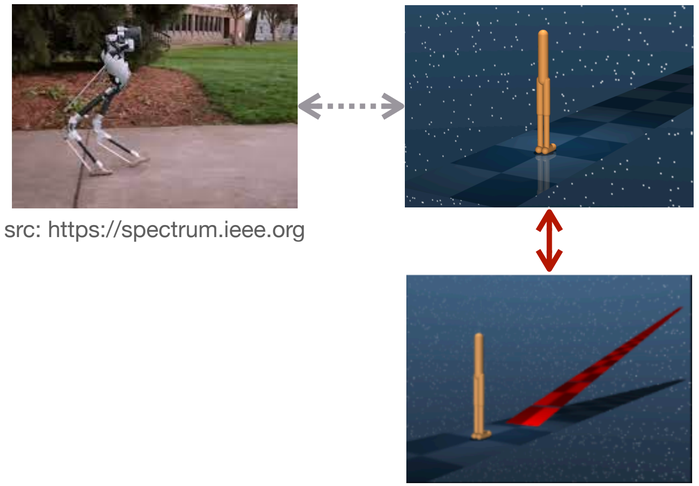
Apple Confidential–Internal Use Only



TM and © 2022 Apple Inc. All rights reserved.

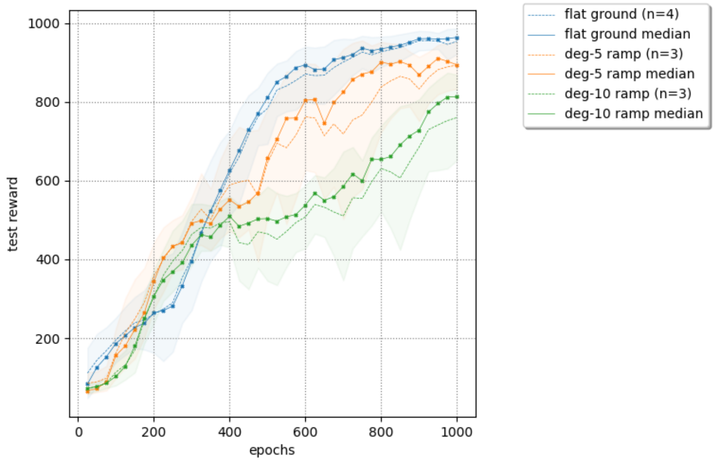
Contact-rich Continuous Control

A key challenge for world models.

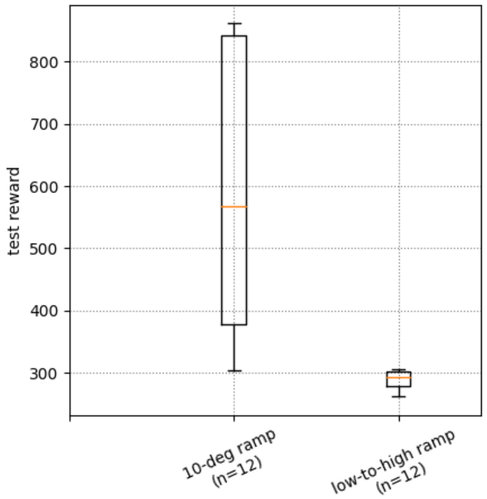


src: https://spectrum.ieee.org

Performance on variable ramp angles



Generalisation to out-of-distribution ramp angles



Underlying Question Can multi-modal supervision improve visual model's test-time performance?

Can learnt contact-aware world models make policy learning more efficient?

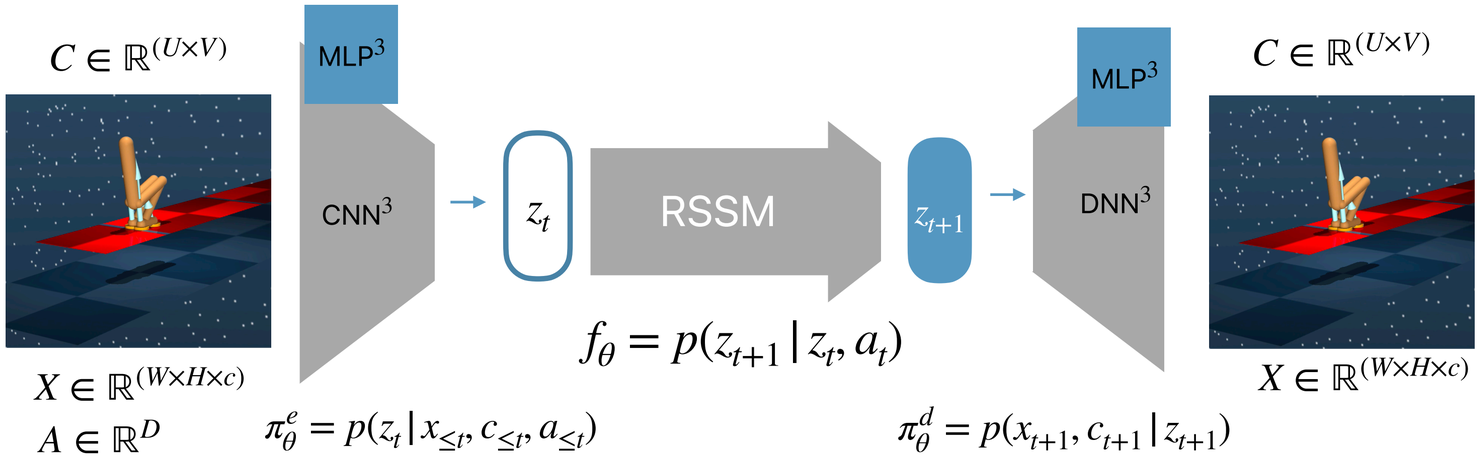
How robust are contact-aware world models to physical variations?

Apple Confidential–Internal Use Only

citations!

Our Approach

A world model trained with contact supervision.



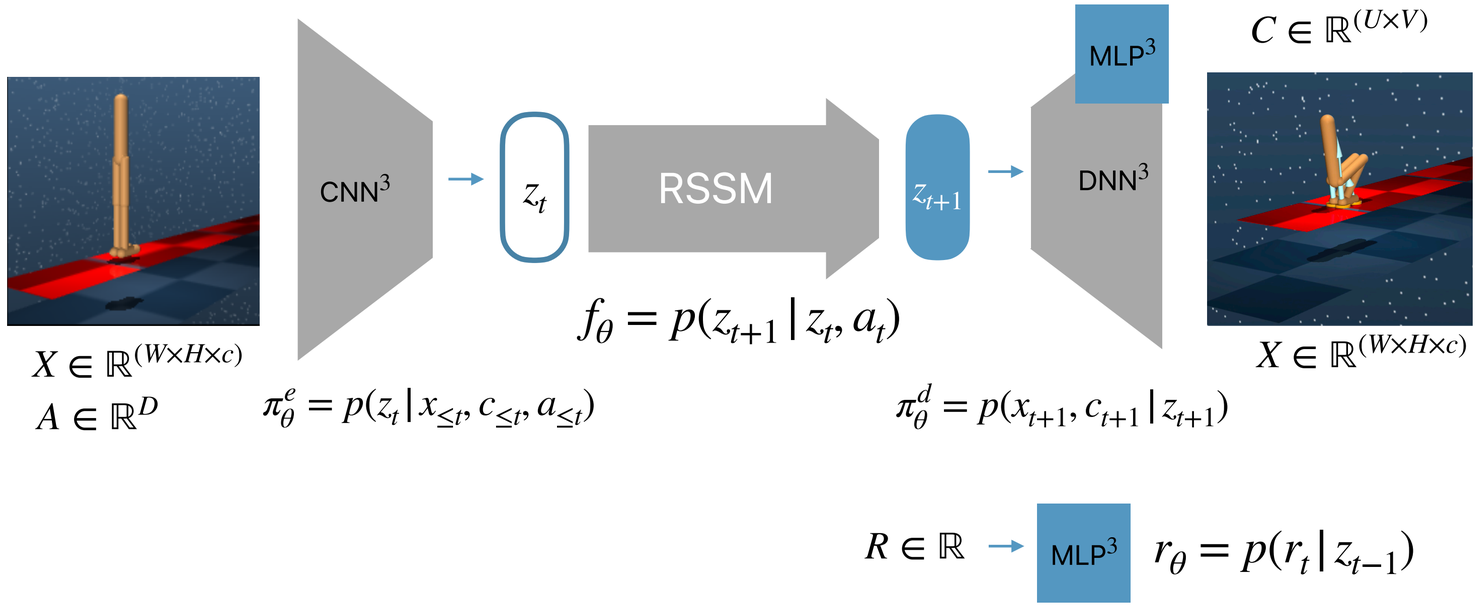
Learn a world model $M_\theta = \{\pi^e_\theta, f_\theta, \pi^d_\theta, r_\theta\}$ that maximises the likelihood of observed trajectory $p(x, c \mid a)$, assuming that $I(X; C) > \delta > 0$. Then optimise a policy $\pi(x, c; M_\theta)$ to maximise expected return.

Apple Confidential–Internal Use Only

citations!

Our Approach

A world model trained with contact supervision.



Learn a world model $M_\theta = \{\pi^e_\theta, f_\theta, \pi^d_\theta, r_\theta\}$ that maximises the likelihood of observed trajectory $p(x, c \mid a)$, assuming that $I(X; C) > \delta > 0$. Then optimise a policy $\pi(x; M_\theta)$ to maximise expected return R.

Apple Confidential–Internal Use Only

Our Approach

A world model trained with contact supervision.

Given a in $A \in \mathbb{R}^D$ and x in $X \in \mathbb{R}^{(W \times H \times C)}$ & c in $C \in \mathbb{R}^{(U \times V)}$, learn a world model $M_\theta = \{\pi^e_\theta, f_\theta, \pi^d_\theta, r_\theta\}$ that maximises the likelihood of observed trajectory $p(x, c \mid a)$, assuming that $I(X; C) > \delta > 0$.

$$\pi^e_\theta = p(z_t \mid x_{\leq t}, a_{\leq t}), f_\theta = p(z_t \mid z_{t-1}, a_{t-1}), \pi^d_\theta = p(x_t, c_t \mid z_t), r_\theta = p(r_t \mid z_{t-1})$$

$$\max_\theta \sum_t \mathsf{E}_{\pi^e_\theta(t)} \left[\ln p(x_t, c_t \mid z_t)\right] - \mathsf{E}_{\pi^e_\theta(t-1)} \left[\mathsf{KL}\left[\pi^e_\theta(t) \,||\, p(z_t \mid z_{t-1}, a_{t-1})\right]\right]$$

$$\max_\theta \sum_t \mathsf{E}_{\pi^e_\theta(t)} \left[\ln p(x_t \mid z_t)\right] - \mathsf{E}_{\pi^e_\theta(t-1)} \left[\mathsf{KL}\left[\pi^e_\theta(t) \,||\, p(z_t \mid z_{t-1}, a_{t-1})\right]\right]$$

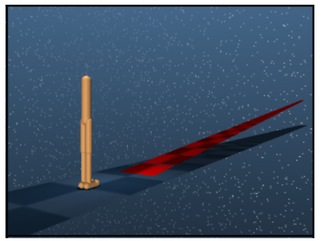
subject to $\|\pi^d(z_t) - c_t\|_2^2 < \epsilon, t \in \{0, \ldots, T-1\}$

Optimise a policy $\pi(x; M_\theta)$ to maximise expected return.

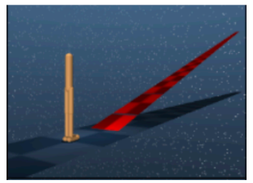
Apple Confidential–Internal Use Only

Key Finding

Contact-aware world models yield policies capable of *zero-shot transfer.*

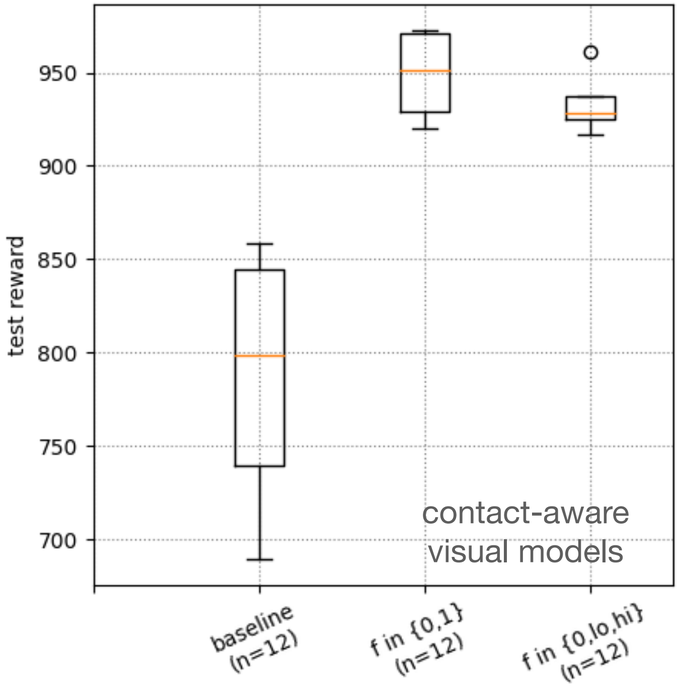


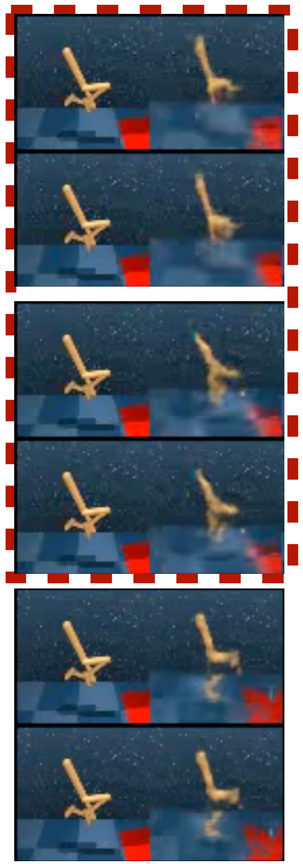
$\theta \sim \text{Unif}(0^\circ, 5^\circ)$



$\theta \sim \text{Unif}(5^\circ, 10^\circ)$

Case 2: generalisation to *unseen* terrains





✓

Apple Confidential–Internal Use Only